The Tell-Tale Cube

Antoine Chédin[1], Matteo Francia[2], Patrick Marcel[1], Verónika Peralta[1], and Stefano Rizzi[2](✉)

[1] LIFAT, University of Tours, Tours, France
[2] DISI, University of Bologna, Bologna, Italy
stefano.rizzi@unibo.it

Abstract. The Intentional Analytics Model (IAM) has been recently envisioned as a new paradigm to couple OLAP and analytics. It relies on two basic ideas: (i) letting the user explore data by expressing her analysis intentions rather than the data she needs, and (ii) returning enhanced cubes, i.e., multidimensional data annotated with knowledge insights in the form of model components (e.g., clusters). In this paper we provide a proof-of-concept for the IAM vision by delivering an end-to-end implementation of describe, one of the five intention operators introduced by IAM. Among the research challenges left open in IAM, those we address are (i) automatically tuning the size of models (e.g., the number of clusters), (ii) selecting the most effective chart or graph for visualizing each cube depending on its features, and (iii) devising a visual metaphor to display enhanced cubes and interact with them.

1 Introduction

Data warehousing and OLAP (On-Line Analytical Processing) have been progressively gaining a leading role in enabling business analyses over enterprise data since the early 90's. During these thirty years, the underlying technologies have evolved from the early relational implementations (still widely adopted in corporate environments), to the new architectures solicited by Business Intelligence 2.0 scenarios, and up to the challenges posed by the integration with big data settings. However, recently, it has become more and more evident that the OLAP paradigm, alone, is no more sufficient to keep the pace with the increasing needs of new-generation decision makers. Indeed, the enormous success of machine learning techniques has consistently shifted the interest of corporate users towards sophisticated analytical applications.

In this direction, the *Intentional Analytics Model* (IAM) has been envisioned as a way to tightly couple OLAP and analytics [20]. The IAM approach relies on two major cornerstones: (i) the user explores the data space by expressing her analysis *intentions* rather than by explicitly stating what data she needs, and (ii) in return she receives both multidimensional data and knowledge insights in the form of annotations of interesting subsets of data. As to (i), five intention operators are proposed, namely, describe (describes one or more cube measures, possibly focused on one or more level members), assess (judges one or more

© Springer Nature Switzerland AG 2020
J. Darmont et al. (Eds.): ADBIS 2020, LNCS 12245, pp. 204–218, 2020.
https://doi.org/10.1007/978-3-030-54832-2_16

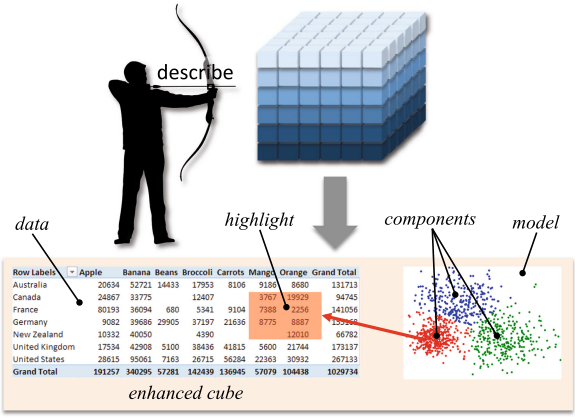

Fig. 1. The IAM approach

cube measures with reference to some baseline), explain (reveals some hidden information in the data the user is observing, for instance in the form of a correlation between two measures), predict (shows data not in the original cubes, derived for instance with regression), and suggest (shows data similar to those the current user, or similar users, have been interested in). As to (ii), first-class citizens of the IAM are *enhanced cubes*, defined as multidimensional cubes coupled with *highlights*, i.e., sets of cube cells associated with interesting components of *models* automatically extracted from cubes [20]. Each operator is applied to an enhanced cube and returns a new enhanced cube. To assess interestingness of model components, a measure based on their (objective) significance and (subjective) surprise is proposed.

An overview of the approach is given in Fig. 1. Noticeably, having different models automatically computed and evaluated in terms of their interestingness relieves the user from the time-wasting effort of trying different possibilities.

Example 1. Let a SALES cube be given, and let the user's intention be

with SALES describe quantity for month ='1997-04' by type using clustering size 3

First, the subset of cells for April 1997 are selected from the SALES cube, aggregated by product type, and projected on measure quantity (in OLAP terms, a slice-and-dice and a roll-up operator are applied). Then, these cells are clustered into 3 clusters based on the values of quantity. Finally, a measure of interestingness is computed for each cluster, and the cells belonging to the cluster with maximum interestingness are highlighted in the results shown to the user. □

Clearly, the IAM vision raises a number of research challenges, e.g., (i) investigate if there are any other intention operators that should be considered besides the basic ones proposed, and how different operators can be combined; (ii) find

techniques for automatically tuning the algorithms that create enhanced cubes by computing models; (iii) enrich the IAM framework with an approach to select the most effective chart or graph for visualizing each cube depending on its features such as number of dimensions, size, etc.; (iv) devise a visual metaphor for displaying enhanced cubes and interacting with them.

The goal of this paper is to provide a proof-of-concept for the IAM vision by delivering an end-to-end implementation of the describe operator. Specifically, we address challenges (ii) by experimenting two techniques to automatically set the number of model components. We also address challenges (iii) and (iv) by proposing a visualization that couples text-based representations and selected graphical representations with a component-driven interaction paradigm. This will save the time it would take to the user to try different visualizations; besides, by automatically selecting the most suitable charts based on the features of each cube, we discourage the user from adopting inappropriate visualizations which might lead her to wrong interpretations of data.

The paper outline is as follows. After introducing a formalism to manipulate cubes and queries in Sect. 2, in Sect. 3 we introduce models, components, and enhanced cubes, and define an interestingness measure. Then, in Sect. 4 we show how an intention is transformed into an execution plan, in Sect. 5 we discuss how to automatically set the model size, i.e., its number of components, and in Sect. 6 we explain how enhanced cubes are visualized. Finally, in Sect. 7 we discuss the related literature, while in Sect. 8 we draw the conclusion.

2 Formalities

To simplify the formalization, we will restrict to consider linear hierarchies (the presence of branches in hierarchies makes Definition 6 overly complex).

Definition 1 (Hierarchy and Cube Schema). *A* hierarchy *is a triple* $h = (L_h, \succeq_h, \geq_h)$ *where:*

(i) L_h *is a set of categorical* levels*, each coupled with a* domain $Dom(l)$ *including a set of* members*;*
(ii) $\succeq_h$ *is a* roll-up *total order of* L_h*; and*
(iii) $\geq_h$ *is a* part-of *partial order of* $\bigcup_{l \in L_h} Dom(l)$*.*

The top level of $\succeq_h$ *is called* dimension*. The part-of partial order is such that, for each couple of levels* l *and* l' *such that* $l \succeq_h l'$*, for each member* $u \in Dom(l)$ *there is exactly one member* $u' \in Dom(l')$ *such that* $u \geq_h u'$*. A* cube schema *is a couple* $\mathcal{C} = (H, M)$ *where:*

(i) H *is a set of hierarchies;*
(ii) M *is a set of numerical measures, with each measure* $m \in M$ *coupled with one aggregation operator* $op(m) \in \{\texttt{sum}, \texttt{avg}, \ldots\}$*.*

Example 2. For our working example it is $\mathsf{SALES} = (H, M)$ where

$$H = \{h_{\mathsf{Date}}, h_{\mathsf{Customer}}, h_{\mathsf{Product}}, h_{\mathsf{Store}}\},\ M = \{\mathsf{quantity}, \mathsf{storeSales}, \mathsf{storeCost}\},$$
$$\mathsf{date} \succeq \mathsf{month} \succeq \mathsf{year},\ \mathsf{customer} \succeq \mathsf{gender}$$
$$\mathsf{product} \succeq \mathsf{type} \succeq \mathsf{category},\ \mathsf{store} \succeq \mathsf{city} \succeq \mathsf{country}$$

and $op(\mathsf{quantity}) = op(\mathsf{storeSales}) = op(\mathsf{storeCost}) = \mathtt{sum}$. □

Aggregation is the basic mechanism to query cubes, and it is captured by the following definition of group-by set. As normally done when working with the multidimensional model, if a hierarchy h does not appear in a group-by set it is implicitly assumed that a complete aggregation is done along h.

Definition 2 (Group-by Set and Coordinate). *Given cube schema $\mathcal{C} = (H, M)$, a* group-by set *of $\mathcal{C}$ is a set of levels, at most one from each hierarchy of H. The partial order induced on the set of all group-by sets of $\mathcal{C}$ by the roll-up orders of the hierarchies in H, is denoted with $\succeq_H$. A* coordinate *of group-by set G is a tuple of members, one for each level of G. Given coordinate γ of group-by set G and another group-by set G' such that $G \succeq_H G'$, we will denote with $rup_{G'}(\gamma)$ the coordinate of G' whose members are related to the corresponding members of γ in the part-of orders, and we will say that γ* roll-ups *to $rup_{G'}(\gamma)$. By definition, $rup_G(\gamma) = \gamma$.*

Example 3. Two group-by sets of SALES are $G_1 = \{\mathsf{date}, \mathsf{type}, \mathsf{country}\}$ and $G_2 = \{\mathsf{month}, \mathsf{category}\}$, where $G_1 \succeq_H G_2$. G_1 aggregates sales by date, product type, and store country (for all customers), G_2 by month and category. Example of coordinates of the two group-by sets are, respectively, $\gamma_1 = \langle$1997-04-15, Fresh Fruit, Italy$\rangle$ and $\gamma_2 = \langle$1997-04, Fruit$\rangle$, where $rup_{G_2}(\gamma_1) = \gamma_2$. □

Definition 3 (Cube). *A* cube *over $\mathcal{C}$ is a tuple $C = (G_C, M_C, \omega_C)$ where:*

(i) G_C is a group-by set of $\mathcal{C}$;
(ii) $M_C \subseteq M$;
(iii) ω_C is a partial function that maps the coordinates of G_C to a numerical value for each measure $m \in M_C$.

Each coordinate that participates in ω_C, with its associated measure values, is called a *cell* of C. With a slight abuse of notation, we will write $\gamma \in C$ to state that γ is a cell of C. A cube whose group-by set G_C includes all and only the dimensions of the hierarchies in H and such that $M_C = M$, is called a *base cube*, the others are called *derived cubes*. In OLAP terms, a derived cube is the result of either a roll-up, a slice-and-dice, or a projection made over a base cube; this is formalized as follows.

Definition 4 (Cube Query). *A* query *over cube schema $\mathcal{C}$ is a triple $q = (G_q, P_q, M_q)$ where:*

(i) G_q is a group-by set of H;

(ii) P_q *is a (possibly empty) set of selection predicates each expressed over one level of* H*;*
(iii) $M_q \subseteq M$*.*

Let C_0 *be a base cube over* $\mathcal{C}$*. The result of applying* q *to* C_0 *is a derived cube* C *such that (i)* $G_C = G_q$*, (ii)* $M_C = M_q$*, and (iii)* ω_C *assigns to each coordinate* $\gamma \in C$ *satisfying the conjunction of the predicates in* P_q *and to each measure* $m \in M_C$ *the value computed by applying* $op(m)$ *to the values of* m *for all the coordinates of* C_0 *that roll-up to* γ*.*

Example 4. The cube query over SALES used in Example 1 is $q = (G_q, P_q, M_q)$ where $G_q = \{\mathsf{type}\}$, $P_q = \{\mathsf{month} = \text{'1997-04'}\}$, and $M_q = \{\mathsf{quantity}\}$. A cell of the resulting cube is ⟨Canned Fruit⟩ with associated value 138 for quantity. □

3 Enhancing Cubes with Models

Models are concise, information-rich knowledge artifacts [19] that represent relationships hiding in the cube cells. The possible models range from simple functions and measure correlations to more elaborate techniques such as decision trees, clusterings, etc. A model is bound to (i.e., is computed over the levels/measures of) one cube, and is made of a set of components (e.g., a clustering model is made of a set of clusters). In the IAM, a relevant role is taken by data-to-model mappings. Indeed, a model partitions the cube on which it is computed into two or more subsets of cells, one for each component (e.g., the subsets of cells belonging to each cluster).

Definition 5 (Model and Component). *A* model *is a tuple* $\mathcal{M} = (t, alg, C, In, Out, \mu)$ *where:*

(i) t *is the model type;*
(ii) alg *is the algorithm used to compute* $\mathcal{M}$
(iii) C *is the cube to which* $\mathcal{M}$ *is bound;*
(iv) In *is the tuple of levels/measures of* $\mathcal{C}$ *and parameter values supplied to* alg *to compute* $\mathcal{M}$*;*
(v) Out *is the set of components that make up* $\mathcal{M}$*;*
(vi) μ *is a function mapping each cell of* C *to one component of* Out*.*

Each model component *is a tuple of a component identifier plus a variable number of properties that describe that component.*

In the scope of this work, it is $t \in \{\text{top-k}, \text{bottom-k}, \text{skyline}, \text{outliers}, \text{clustering}\}$. The components for these model types are as follows:

1. For t = top-k, there are two components: one for top-k cells, one for the others (similarly for bottom-k). Each component is described by the average z-score of its cells.

2. For t = skyline, there are two components: one for the cells in the skyline, one for the others.
3. For t = outliers, there are two components: one for outlier cells, one for the others. Each component is described by its outlierness.
4. For t = clustering, there is one component for each cluster. Each component is described by the centroid of the corresponding cluster.

Example 5. A possible model over the SALES cube is characterized by

$$\begin{aligned}
&t = \text{clustering},\ alg = \text{K-Means},\ C = \mathsf{SALES},\\
&In = \langle \mathsf{quantity}, n = 5, rndSeed = 0\rangle,\ Out = \{c1, \ldots, c5\},\\
&\mu(\langle \text{Bagels}\rangle) = \mu(\langle \text{Batteries}\rangle) = \mu(\langle \text{Wine}\rangle) = \mu(\langle \text{Beer}\rangle) = c1;\\
&\mu(\langle \text{Canned Fruit}\rangle) = \mu(\langle \text{Muffins}\rangle) = \mu(\langle \text{Fresh Fruit}\rangle) = c2;\ \ldots
\end{aligned}$$

where n is the desired number of clusters and $rndSeed$ is the seed to be used by the k-means algorithm to randomly generate the 5 seed clusters. Component $c1$ is characterized by property *centroid* with value 151.3. □

As the last step in the IAM approach, the cube is enhanced by associating it with a highlight, i.e., with the subset of cells corresponding to the most interesting component of the model; these cells are determined via function μ. The measure proposed in [20] to asses the interestingness of each component is based on the idea of *prior belief* [3]: specifically, it defines the *surprise* of a component as the difference of belief for corresponding cells in the cube before (C) and after (C') the application of the intention. This requires first of all to define the concept of "corresponding cell(s)" in C of each cell γ' of C', which is done through a *proxy* function. To simplify this function, we will assume that the intention does not change both the cube group-by set and its selection predicate. Intuitively, if the intention changes the group-by set, the corresponding cell(s) of γ' are determined via the part-of order; if the intention changes the selection predicate, the corresponding cell of γ' is γ' itself if it is part of C, the whole cube C otherwise.

Definition 6 (Proxy Cells). *Let C and C' be two cubes over cube schema $\mathcal{C}$, and let q and q' the queries producing C and C', respectively. Let γ' be a coordinate of C'. The* proxy cells *of γ' over C are defined as*

$$proxy(\gamma') = \begin{cases} \{\gamma : rup_{G_{q'}}(\gamma) = \gamma'\},\ \textit{if } G_q \succ_H G_{q'} \\ \{rup_{G_q}(\gamma')\},\ \textit{if } G_{q'} \succ_H G_q \end{cases},\ \textit{if } G_{q'} \neq G_q\ (P_{q'} = P_q);$$

$$proxy(\gamma') = \begin{cases} \{\gamma'\},\ \textit{if } \gamma' \in C \\ \{\gamma : \gamma \in C\},\ \textit{if } \gamma' \notin C \end{cases},\ \textit{if } P_{q'} \neq P_q\ (G_{q'} = G_q);$$

$$proxy(\gamma') = \{\gamma'\},\ \textit{otherwise.}$$

For the first intention, the starting cube C is the base cube; since in this case the user has no prior belief, we put $proxy(\gamma') = \varnothing$ for all $\gamma' \in C'$.

Definition 7 (Interestingness). *Let* $\mathcal{M} = (t, alg, C', In, Out, \mu)$ *be a model and* $c \in Out$ *be one of its components. The* interestingness *of* c *is defined as*

$$int(c) = avg_{\gamma' \in \mu^{-1}(c)}\{surprise(\gamma')\}, \text{ where}$$
$$surprise(\gamma') = max_{m \in C'}\{z_m(\gamma') - avg_{\gamma \in proxy(\gamma')}\{z_m(\gamma)\}\}$$

and function $z_m()$ *returns the z-score of a cell for measure* m *over the whole cube that cell belongs to.*

Definition 8. *An* enhanced cube E *is a triple of a cube* C*, a set of models* $\{\mathcal{M}_1, \ldots, \mathcal{M}_r\}$ *bound to* C*, and a highlight* $c_{high} = argmax_{\{c \in \bigcup_{i=1}^{r} Out_i\}}(int(c))$.

Example 6. Let

with SALES describe quantity for month = '1997-04' by type
with SALES describe quantity for month = '1997-04' by category

be a sequence of two intentions formulated by the user. While the plan generated for the first intention relies on query q defined in Example 4, the one for the second intention relies on $q' = (G_{q'}, P_{q'}, M_{q'})$ where $G_{q'} = \{\text{category}\}$, $P_{q'} = \{\text{month} = \text{'1997-04'}\}$, and $M_{q'} = \{\text{quantity}\}$. Let C and C' be the cubes resulting from q and q', respectively. In practice, q' corresponds to a roll-up of q, so the proxy cells of each cell in C' are determined using the *rup* operator as shown in Fig. 2 (red lines). The figure also shows the z-score for each cell of C and C', and the surprise of each cell of C'. Then, let $\mathcal{M}$ be the model of type top-k, with $k = 1$, computed on C'; this model has two components: c_1, including only the top-1 cell, and c_2, including all the others (shown in green and yellow in the figure). The interestingness values for these two components are $int(c_1) = 1.5$ and $int(c_2) = 0.8$, respectively. So, the enhanced cube resulting from the second intention includes C', $\mathcal{M}$, and the highlight c_1. □

type	quantity	*z-score*
Bagels	48	-1,0
Beer	116	-0,6
Bologna	192	-0,2
Canned Fruit	138	-0,5
Deli Meats	211	-0,1
Fresh Chicken	64	-0,9
Fresh Fruit	798	3,0
Frozen Chicken	237	0,0
Hamburger	141	-0,5
Hot Dogs	154	-0,4
Muffins	205	-0,1
Sliced Bread	266	0,2
Wine	448	1,1

category	quantity	*z-score*	*surprise*
Beer and Wine	564	-0,9	-1,1
Bread	519	-1,1	-0,8
Fruit	936	0,8	-0,4
Meat	999	1,1	1,5

Fig. 2. Cubes C (left) and C' (right) in Example 6; in green and yellow two components for the top-k model, in red the proxy relationship for two cells in C' (Color figure online)

4 Execution Plans for Describe intentions

The describe operator provides an answer to the user asking "show me my business" by describing one or more cube measures, possibly focused on one or more level members, at some given granularity [20]. The cube is enhanced by showing either the top/bottom-k cells, the skyline, the outliers, or clusters of cells.

The basic idea we followed to define the syntax for describe is that the user will work in sessions, similarly to the OLAP paradigm. As a consequence, the group-by set (by clause) and the selection predicate (for clause) of each intention are formulated as increments over those of the previous intention.

Let C be a cube returned by executing query $q = (G_q, P_q, M_q)$ over base cube C_0 having cube schema $\mathcal{C} = (H, M)$. The general syntax for describe is

$$\text{with } C \text{ describe } m_1, \ldots, m_z \text{ [for } P' \text{] [by } l' \text{] [using } t_1, \ldots, t_r \text{] [size } k \text{]}$$

where $m_1, \ldots, m_z \in M$ are measures of $\mathcal{C}$, P' is a set of selection predicates each over one level of H, l' is a level of H, $t_1, \ldots, t_r$ are model types, and k is the desired size to be applied to all the models returned as explained in point 2 below (optional parts of the syntax are in brackets).

Now let l be the level in G_q belonging to the same hierarchy of l', and let $P \subseteq P_q$ be the set of predicates in P_q expressed over levels belonging to the same hierarchies than those over which the predicate in P' are expressed. The plan corresponding to a fully-specified intention with C, i.e., one where all optional clauses have been specified, is:

1. Execute query $q' = (G_{q'}, P_{q'}, \{m_1, \ldots, m_z\})$, where $G_{q'} = G_q \setminus \{l\} \cup \{l'\}$ and $P_{q'} = P_q \setminus P \cup P'$. Let C' be the cube resulting from the execution of q' over C_0.
2. For $1 \leq i \leq r$, compute model $\mathcal{M}_i = (t_i, alg_i, C', In_i, Out_i, \mu_i)$ and for each $c \in Out_i$, compute $int(c)$. Size k is used for clustering to determine the number of clusters to be computed, for top-k and bottom-k to determine the number of cells to be returned, for outliers to determine the number of outliers; it is neglected for the skyline.
3. Find the highlight $c_{high} = argmax_{\{c \in \bigcup_i Out_i\}}(int(c))$.
4. Return the enhanced cube consisting of C', $\{\mathcal{M}_1, \ldots \mathcal{M}_r\}$, and highlight c_{high}.

Partially-specified intentions are interpreted as follows:

- If the for P' clause has not been specified, we consider $P_{q'} = P_q$.
- If the by l' clause has not been specified, we consider $G_{q'} = G_q$.
- If the using $t_1, \ldots, t_r$ clause has not been specified, all model types listed in Sect. 3 are computed over C' (the skyline is computed only if $z > 1$, i.e., at least two measures have been specified).
- If the size k clause has not been specified, the value of k is determined automatically as discussed in Sect. 5.

When the first intention is issued on a cube, to constrain the size of the data to be returned it is required that either the by or the for clauses are specified at least.

Example 7. Consider the following session on the SALES cube:

with SALES describe quantity for month = '1997-04' by type
with SALES describe quantity by category using clustering size 3
with SALES describe quantity, storeSales for country = 'Italy' using skyline

The queries corresponding to the first two intentions are q in Example 6 and q' in Example 6, respectively. The query corresponding to the third intention is q'' characterized by $G_{q''} = \{\mathsf{category}\}$, $P_{q''} = \{\mathsf{month} = \text{'1997-04'}, \mathsf{country} = \text{'Italy'}\}$, and $M_{q''} = \{\mathsf{quantity}, \mathsf{storeSales}\}$. The models computed for the first intention are top-k, bottom-k, clustering, and outliers (computing the skyline for a single measure makes no sense). For the second and the third intentions, a clustering producing 3 clusters and the skyline are computed, respectively. □

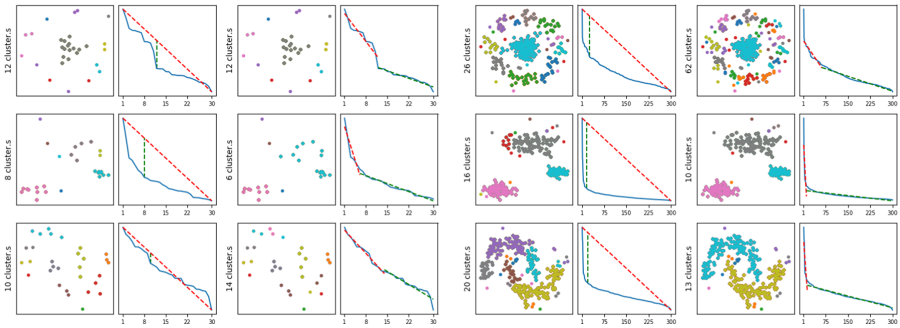

Fig. 3. Results on the 30 (left) and 300 (right) samples data for Kneedle (columns 1 and 3) and L-method (columns 2 and 4)

5 Setting the Model Size

Our approach to find the best value for the size parameter k when it is not specified in the intention is based on good practices in hierarchical clustering, especially when single-linkage is used, meaning that inter-cluster distance is measured by the closest two points of the clusters. The best separation of clusters can then be found by finding the knee of the evaluation graph of the clustering algorithm, which is a two dimensional plot where the x-axis is the number of clusters produced and the y-axis is one classical clustering evaluation metric (error, silhouette, etc.) considering x clusters. In hierarchical clustering, since

the cost for merging clusters constantly increases, the evaluation graph often looks like an L-shaped curve with a more or less defined knee. The assumption usually made is that the best merging cost threshold is at the curve knee, where the curve switches from a sharp slope to a low decreasing line.

We tested two solutions from the literature, namely L-method [15] and Kneedle [18], which have been proposed to find the knee in a curve of discrete data. These methods were compared using 3-dimensional non-random toy datasets specifically created for the experiment with the *Scikit-Learn* Python package, varying the size (6, 30, and 300 samples) and the shape of clusters, defining a ground truth. Due to lack of space, we only report the main findings. The code for generating the datasets and the tests is available at https://github.com/antoinechedin/descriptive-clustering.

While both methods achieve similar good results for knee detection, the L-method takes longer to execute and tends to shift the knee on large data sets. This can be seen, for instance, in Fig. 3 on the top right graph. The right knee seems to be located at $x = 25$ but the method returned a knee at $x = 62$. Since Kneedle is quicker and provides more consistent results, we have adopted it to determine k, both for clustering (k being the number of clusters), top/bottom-k (where k is the number of points in the first cluster, i.e., the one with higher values) and outliers (where k is the number of points in the first and last cluster).

6 Visualizing Enhanced Cubes

In this section we discuss how to provide an effective description of an enhanced cube by coupling text-based representations (a pivot table and a ranked component list) and graphical representations (one or more charts) with an ad-hoc interaction paradigm. The guidelines we adopt to this end are explained below:

(i) For visualization purposes, we assume that an intention can select at most three measures ($1 \leq z \leq 3$) and three group-by levels ($1 \leq n \leq 3$). This is actually not a strong limitation, considering that a visualization of four or more dimensions and/or measures using a single table or chart is hardly feasible and definitely not intuitive.

(ii) Since we are focusing on intentions aimed at *describing* data, we believe that providing multiple visualizations from different points of view should be preferred to just picking the "most effective one". Indeed, the effectiveness of a visualization type largely depends on the skills and personal tastes of each user.

(iii) We restrict to considering visualization types that can be easily understood both by lay users and skilled users, and are suitable for multidimensional data.

(iv) Clearly, the effectiveness of a visualization type also depends on the features of the specific dataset. Using an unsuitable visualization can generate confusion and misunderstandings in users, and can lead them to wrong conclusions. Thus, for each intention we visualize only the charts that are recognized to be suitable given the characteristics of the data to be shown.

(v) Models and components play a key role in the IAM approach. Thus, the visualizations we provide aims at showing not only dimension and measure values, but also the different components of a model using a color code. For the same reason, the interaction paradigm should be component-driven.

The visualization we provide for enhanced cube E based on guidelines (ii) and (v) includes three distinct but inter-related areas: a *table* area that shows the cube cells using a pivot table; a *chart* area that complements the table area by representing the cube cells through one or more charts; a *component* area that shows a list of model components sorted by their interestingness. The chart types we consider following guidelines (i) and (iii) are multiple line graphs, grouped column charts, heat maps, bubble graphs, and scatter plots. The heuristics we adopt to decide whether using or not each chart type for a given enhanced cube E (guideline (iv)) was inspired by [10]. The features of E we take into account to this end are the number n of dimensions, the number z of measures, and the domain cardinality and type of the dimensions. The pseudocode is shown in Algorithm 1; an example of heuristics we use is: if $n = 2$ and $z = 1$, draw a bubble graph using the X and Y axes for the two dimensions, the bubble size for the measure, and the bubble color for the model components.

The interaction paradigm we adopt is component-driven (guideline (v)). Specifically, clicking on one component c in the component area leads to emphasizing the corresponding cube cells (i.e., those that map to c via function μ) both in the table area and in the chart area. The highlight is the top component in the list and is selected by default.

Algorithm 1. Chart area creation

Require: $D = \{d_1, \ldots, d_n\}$: sets of dimensions, $M = \{m_1, \ldots, m_z\}$: set of measures, $T = \langle t_1, \ldots, t_r \rangle$: list of models sorted by interestingness

```
if n = 1 and isDate(d_1) then                                   ▷ Visualize multiple line graph
    MultipleLineGraph(X : d_1, Y_1 : m_1, ..., Y_z : m_z)
if z = 1 and card(d_1) ≤ 50 then                                ▷ Visualize grouped column chart
    if n = 1 then
        GroupedColumnChart(X : d_1, height : m_1, color : t_1)
    else
        if n = 2 and card(d_2) ≤ 8 then
            GroupedColumnChart(X : d_1, height : m_1, color : d_2)
if n = 2 and z = 1 then                                         ▷ Visualize heat map
    HeatMap(X : d_1, Y : d_2, color : m_1)
else
    if n = 1 then
        HeatMap(X : d_1, Y_1 : m_1, ..., Y_z : m_z)
if z = 1 then                                                   ▷ Visualize bubble chart
    switch n do
        case 2
            BubbleChart(X : d_1, Y : d_2, size : m_1, color : t_1)
        case 3
            BubbleChart(X : d_1, Y : d_2, Z : d_3, size : m_1, color : t_1)
switch z do                                                     ▷ Visualize scatter plot
    case 2
        ScatterPlot(X : m_1, Y : m_2, color : t_1)
    case 3
        ScatterPlot(X : m_1, Y : m_2, Z : m_3, color : t_1)
```

Example 8. Figure 4 shows the visualization obtained when a session of the two following intentions is formulated:

with SALES describe storeCost by month

with SALES describe storeCost by category

On the top left, the table area; on the right, the chart area; on the bottom left, the component area (the three areas have been repositioned in the figure to save space). Here it is $n = 2$ and $z = 1$, so a heat map and a bubble chart have been selected. The top-interestingness component is a cluster, so a color has been assigned to each component of clustering (i.e., to each cluster) and is uniformly used in all three areas. The highlight (in green) is currently selected and is emphasized using a thicker border in all areas. Note that a tooltip with all the details about a single cell is also shown (in yellow).

7 Related Work

The idea of coupling data and analytical models was born in the 90's with inductive databases, where data were coupled with patterns meant as generalizations of the data [14]. Later on, data-to-model unification was addressed in MauveDB

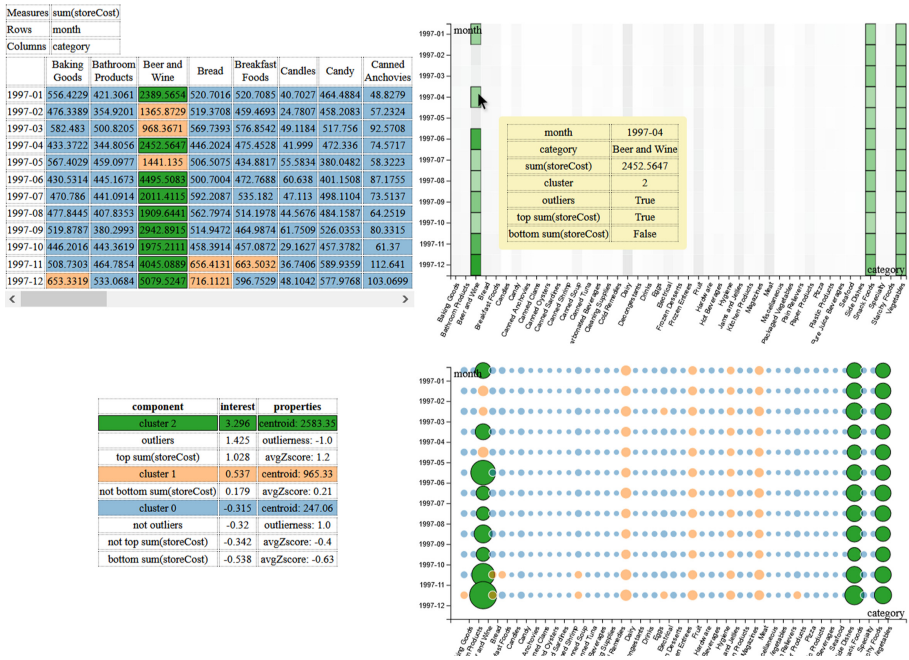

Fig. 4. The visualization obtained for the session in Example 8 (Color figure online)

[7], which provides a language for specifying model-based views of data using common statistical models. However, achieving a unified view of data and models was still seen as a research challenge in business intelligence a few years later [13]. More recently, Northstar [12] has been proposed as a system to support interactive data science by enabling users to switch between data exploration and model building, adopting a real-time strategy for hyper-parameter tuning. Finally, the coupling of data and models is at the core of the IAM vision [20], on which this paper relies. The three basic pillars of IAM are (i) the redefinition of query as expressing the user's intention rather than explicitly declaring what data are to be retrieved, (ii) the extension of query results from plain data cubes to cubes enhanced with models and highlights, and (iii) the characterization of model components in terms of their interestingness to users.

The coupling of the OLAP paradigm and data mining to create an approach where concise patterns are extracted from multidimensional data for user's evaluation, was the goal of some approaches commonly labeled as OLAM [11]. In this context, k-means clustering is used in [2] to dynamically create semantically-rich aggregates of facts other than those statically provided by dimension hierarchies. Similarly, the shrink operator is proposed in [9] to compute small-size approximations of a cube via agglomerative clustering. Other operators that enrich data with knowledge extraction results are DIFF [16], which returns a set of tuples that most successfully describe the difference of values between two cells of a cube, and RELAX [17], which verifies whether a pattern observed at a certain level of detail is also present at a coarser level of detail, too. Finally, in [6] the OLAP paradigm is reused to explore prediction cubes, i.e., cubes where each cell summarizes a predictive model trained on the data corresponding to that cell.

The characteristics of the different approaches for visualizing data and interacting with them have been deeply explored in the literature, also with reference to their suitability for datasets with different features and users with varying skills and goals. In [4], the author surveys the classifications proposed in the literature for visualization types and integrates them into a single comprehensive framework. Abela [1] proposes a decision tree to select the best visualization according to the user's goal and to the main features of data. More recently, SkyViz —to which our approach is inspired— starts from a visualization context based on seven coordinates for assessing the user's objectives and describing the data to be visualized. Then it uses skyline-based techniques to translate a visualization context into a set of suitable visualization types and to find the best bindings between the columns of the dataset and the graphical coordinates used by each visualization type.

8 Evaluation and Conclusion

In this paper we have proposed a visual metaphor to display enhanced cubes for describe intentions according to the IAM vision. The prototype implementation can be accessed at http://semantic.csr.unibo.it/describe/. It uses the simple multidimensional engine described in [8], which in turn relies on the MySQL

DBMS to execute queries on a star schema based on multidimensional metadata (in principle, the prototype could work on top of any other multidimensional engine). The mining models are imported from the Scikit-Learn Python library. Finally, the web-based visualization is implemented in JavaScript and exploits the D3 library for chart visualization.

To evaluate the feasibility of our approach from the computational point of view, we made some preliminary performance tests; to this end we populated the SALES cube using the FoodMart data (https://github.com/julianhyde/foodmart-data-mysql). We considered the worst case, in which all five models are computed on cubes obtained by progressively including in the group-by set the three dimensions with highest cardinality. The tests were run on an Intel(R) Core(TM)i7-6700 CPU@3.40 GHz CPU with 8 GB RAM. Table 1 shows the time necessary to query the base cube, to compute the models, to measure the interestingness, and to generate the pivot table returned to the browser. Remarkably, it turns out that at most 6.38 s are necessary to retrieve and visualize an enhanced cube of 86829 cells.

The main directions for future research we wish to pursue are: (i) evaluate the effectiveness of the approach by experimenting it with real users; (ii) extend the approach to the other intention operators and to operate with *dashboards* of enhanced cubes; and (iii) experiment other interestingness metrics [5].

Table 1. Execution time in seconds for increasing cube cardinalities

Cardinality	Query	Model	Interestingness	Pivot	Total
323	0.88	1.45	0.03	0.03	2.39
77832	0.64	3.61	0.39	0.51	5.14
86829	0.69	3.66	0.48	1.56	6.38

References

1. Abela, A.: Advanced Presentations by Design. Pfeiffer, Hoboken (2008)
2. Bentayeb, F., Favre, C.: RoK: roll-up with the k-means clustering method for recommending OLAP queries. In: Bhowmick, S.S., Küng, J., Wagner, R. (eds.) DEXA 2009. LNCS, vol. 5690, pp. 501–515. Springer, Heidelberg (2009). https://doi.org/10.1007/978-3-642-03573-9_43
3. Bie, T.: Subjective interestingness in exploratory data mining. In: Tucker, A., Höppner, F., Siebes, A., Swift, S. (eds.) IDA 2013. LNCS, vol. 8207, pp. 19–31. Springer, Heidelberg (2013). https://doi.org/10.1007/978-3-642-41398-8_3
4. Börner, K.: Atlas of Knowledge: Anyone Can Map. MIT Press, Cambridge (2015)
5. Chanson, A., Crulis, B., Drushku, K., Labroche, N., Marcel, P.: Profiling user belief in BI exploration for measuring subjective interestingness. In: Proceedings of DOLAP (2019)

6. Chen, B., Chen, L., Lin, Y., Ramakrishnan, R.: Prediction cubes. In: Proceedings of VLDB, pp. 982–993 (2005)
7. Deshpande, A., Madden, S.: MauveDB: supporting model-based user views in database systems. In: Proceedings of SIGMOD, pp. 73–84 (2006)
8. Francia, M., Gallinucci, E., Golfarelli, M.: Towards conversational OLAP. In: Proceedings of DOLAP, pp. 6–15 (2020)
9. Golfarelli, M., Graziani, S., Rizzi, S.: Shrink: an OLAP operation for balancing precision and size of pivot tables. Data Knowl. Eng. **93**, 19–41 (2014)
10. Golfarelli, M., Rizzi, S.: A model-driven approach to automate data visualization in big data analytics. Inf. Vis. **19**(1), 24–47 (2020)
11. Han, J.: OLAP mining: integration of OLAP with data mining. In: Proceedings of Working Conference on Database Semantics, pp. 3–20 (1997)
12. Kraska, T.: Northstar: an interactive data science system. PVLDB **11**(12), 2150–2164 (2018)
13. Pedersen, T.B.: Warehousing the world: a vision for data warehouse research. In: Kozielski, S., Wrembel, R. (eds.) New Trends in Data Warehousing and Data Analysis. Annals of Information Systems, vol. 3, pp. 1–17. Springer, Boston (2009). https://doi.org/10.1007/978-0-387-87431-9_1
14. Raedt, L.D.: A perspective on inductive databases. SIGKDD Explor. **4**(2), 69–77 (2002)
15. Salvador, S., Chan, P.: Determining the number of clusters/segments in hierarchical clustering/segmentation algorithms. In: Proceedings of ICTAI, pp. 576–584 (2004)
16. Sarawagi, S.: Explaining differences in multidimensional aggregates. In: Proceedings of VLDB, pp. 42–53 (1999)
17. Sathe, G., Sarawagi, S.: Intelligent rollups in multidimensional OLAP data. In: Proceedings of VLDB, pp. 531–540 (2001)
18. Satopaa, V., Albrecht, J.R., Irwin, D.E., Raghavan, B.: Finding a "kneedle" in a haystack: Detecting knee points in system behavior. In: Proceedings of ICDCS, pp. 166–171 (2011)
19. Terrovitis, M., et al.: Modeling and language support for the management of pattern-bases. Data Knowl. Eng. **62**(2), 368–397 (2007)
20. Vassiliadis, P., Marcel, P., Rizzi, S.: Beyond roll-up's and drill-down's: an intentional analytics model to reinvent OLAP. Inf. Syst. **85**, 68–91 (2019)